North America Product Development

CUSTOM APPLICATION SPECIFIC
PPEI ELECTRICAL REQUIREMENTS
(CASPER)

2017
ISUZU N-SERIES

June 24, 2015

MY' 2017 LCF

Gen 4 V8	**L96 Engine**	**E78 ECM**
	MYD Auto trans.	**T43 TCM**

GM NAPD Electrical Powertrain Interface

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1.0 INTRODUCTION

1.1 SCOPE OF DOCUMENT

This specification establishes the General Motors Platform to Powertrain Electrical Interface (PPEI) requirements for the E78 engine control module (ECM), and T43 transmission control module (TCM). Components supplied to this specification shall meet all requirements defined or referenced herein unless specifically waived in writing by GM. Components, which do not meet the requirements defined or referenced herein, are subject to rejection.

It is possible and may be beneficial, that at a later date, both parties may modify sections of this specification. Parties to the Isuzu N-Series agreement are therefore encouraged to provide feedback on the effectiveness and completeness of the final document. All feedback to this specification shall be provided in writing to the release engineer at GM.

This specification should not prevent parties to the Isuzu N-Series agreement from suggesting alternative approaches that would result in an improved or more cost-effective product. The parties to the Isuzu N-Series agreement are encouraged to pursue such changes with GM on a case-by-case basis. The parties to the Isuzu N-Series agreement shall present these recommendations or suggested enhancements to the GM release engineer in writing.

The supplier shall not change internal design (including component value/tolerances), sources of purchased parts or processes without prior approval from the Design Release Engineer at GM.

1.2 MISSION / THEME

The information contained in this specification should define the Platform to Powertrain Electrical Interface (PPEI) requirements for the ECM and TCM. This specification defines the platform functions, performance, reliability, and validation requirements for the N-Series.

1.3 CLASSIFICATION

Not Applicable.

1.4 DOCUMENT CONTROL

The information contained in this specification is considered to be GM Confidential. ‘GM Confidential’ is a designation of information, which has competitive value that, if disclosed, would have adverse consequences to GM. As such, GM wishes to protect this information from disclosure to third parties (i.e., persons outside of GM without a ‘need to know’). The GM Information Security Practices and Procedures relating to the handling of GM Confidential materials shall be followed.

1.4.1 DOCUMENT FORMAT

Sections, paragraphs, and sub-paragraphs of this specification will be uniquely numbered as a subset of the category and sub-category of the section into which they fall. Each lower level requirement will bear the same number as its higher-level requirement and will be uniquely identified by the addition of necessary decimal places in the numbering scheme.

1.4.2 CHANGE AUTHORITY

The GM Design System Engineer shall make changes to the format and content of this specification.

1.4.3 DEFINITIONS

The word “shall,” will be used to state binding requirements of the component defined by this document. These requirements will be verifiable.

The word “must,” will state requirements of other components and/or subsystems whose definitions are outside the scope this document.

The word “will” is used to state either of the following:

(1) Conditions that result from immutable laws of physics.

(2) Conditions that result from adherence to other stated, binding requirements.

The words “are” and “is” will state definitions and facts that do not require verification.

2.0 APPLICABLE DOCUMENTS

The following documents form a part of this specification to the extent specified herein. Unless otherwise noted, this specification refers to the most recent version of a particular document.

2.1 PRECEDENCE ORDER

In the event of a conflict between a requirement within this specification and a requirement in a document referred to herein, the requirement within this specification shall take precedence unless it fails to meet or exceed an applicable law or regulation. GM will review those cases where a requirement fails to meet or exceed an applicable law or regulation, and approve any changes to the specification.

Those cases where a conflict occurs between requirements in documents referred to within this specification will be reviewed by GM and changes to the specification shall be approved by GM.

Specific departures from this order of precedence shall be identified in the appropriate paragraph.

2.2 GENERAL MOTORS DOCUMENTS

2.2.1 IVED POWERTRAIN CONTROLLER MECHANIZATION'S (ECM AND TCM)

IVED mechanizations are revised regularly and can be viewed and downloaded at the following website:

http://ived.gm.com/cgi-bin/query_form_schem_pdf.sh

2017 L96, EC01 Gen IV SB with E78 for G

10MY TCM Mech.pdf

Note: Confirm the most recent release.

Isuzu schematics are attached below.

TBD

Note: Confirm the most recent release.

2.2.2 GENERALLY AVAILABLE DOCUMENTS

Contacting Global Engineering Documents at 1-800-624-3974 may obtain the following documents.

The documents can be located at the following websites:

http://pai.gm.com/Global/Eng/EDM/Search.nsf/StandardsSearch_D?OpenForm

http://www.gmsupplypower.com/apps/supplypower/NASApp/spcommon/SPMain.jsp

http://issapp00.naeng.gm.com/sharedrive/

Specifications	
PIM Specification	**Isuzu PIM (Platform Interface Module) Specification** **(Available from Isuzu Engineering)** **Reference EVCS-552 revision 1.1 (June 17, 2015)**
GM6241M	**Labels, Parameter for Underhood and Protected Exterior**
GM6090M (B4-A)	**Flammability Component Test Procedure**
GMW3091 Rev H	**EMC Vehicle Requirement Specifications (April 2006)**
GMW3094 Rev 3	**EMC Vehicle Verification Specifications (August 2001)**
GMW3104 Rev 1.5	**GMLAN Strategy Specification.**
GMW3110 v.E	**Enhanced Diagnostics Test Mode Specification.**

Specifications	
GMW3113 v1.1	**Vehicle Diagnostic and Programming Requirements.**
GM8762 to GM8781 v8.4	**Standard Platform–Powertrain Electrical Interface** PPEI_GenInfoElectricalSerialData_REQ_8g PPEI_AirConditioningCompressorControl_REQ Isuzu 11AU PPEI_PowerAndGround_REQ Isuzu 11A PPEI_BrakesAndTractionControl_REQ Isu PPEI_CruiseControl_REQ Isuzu 11AU14.p PPEI_Transmission_REQ Isuzu 11SE14.p PPEI_PostCollisionOperation_REQ Isuzu PPEI_CoolingFanControl_REQ 8_4 Isuzu PPEI_SerialDataArchitecture_REQ_8.4 Isu NOTE: Excerpts of relevant sections of PPEI v8.4 for Isuzu as determined by program/feature assessment CASPER lite activity. N/A: GM8764, GM8766, GM8767, GM8768, GM8774, GM8775, GM8776, GM8778, GM8780, and GM8781. Changes N/A per document compare (reference PPEI v3.6): GM8765.
GMW3172 RevB	**General Specification for Electrical/Electronic Component & Durability Tests.**
GMW4710GS v1.1	**Technical Specification for Electronic Control Module Traceability and Verification Requirements**
GMSEA156	**Service Programming Symbol (required for new programs)**

Guidelines	
GM 3640	**Serviceability Design Best Practices - GM handbook describing design methods and techniques to ensure that the operation of servicing GM vehicles is an efficient one.**
GM 1644	**General Motors Targets for Excellence - GM handbook describing the degrees of excellence which all GM products should strive for.**

2.2.3 GM INTERNAL USE DOCUMENTS

GM INTERNAL USE DOCUMENTS	
Electric Power Management Spec rev K	**Electric Power Management – Gen IV Subsystem Technical Specification (SSTS).**
Revision 35267.I	**GM Bill of Process for Electrical Programming and Final Line Testing.**
Revision 5.3	GMNAE Power Mode and Vehicle Electrical Power Moding Specification using GMLAN communication Strategy
Revision 7.02	**System Technical Specification for the Common Architecture GMLAN Serial Communication Systems**
Version 7.02	**Common Architecture List Report**
TL.07.0054.R05	**GM WW Software and Calibration Parts API Specification.**
TL.07.0098.R04	**GM Controller Programming Manufacturing Requirements**
TX169879 v1.0	**Requirements for CAL-DS Compatibility**
TL.17.1114.R3.2.1	**Engine Control Module – Transmission Control Module Electrical Interface**
TL.50.1116.R01	**Powertrain Control Module Common PCM Package 4 Vehicle Mounting Requirements**

2.3 INDUSTRY DOCUMENTS

The following documents may be obtained by contacting the Society of Automotive Engineers (SAE) at 412-776-4841.

The documents can be located at the following website:

http://www.sae.org/

Recommended Practices	
SAE J1113	**Electromagnetic Susceptibility Measurement Procedures for Vehicle Components (Except Aircraft).**
SAE J1211	**Recommended Environmental Practices for Electronic Equipment (November 1978).**
SAE J1963	**Diagnostic Connector Status and Pin Usage.**
SAE J1979	**E/E Diagnostic Test Modes.**
SAE J2190	**Enhanced E/E Diagnostic Test Modes.**
SAE J2284	**High Speed CAN for Passenger Vehicle Applications.**

3.0 REQUIREMENTS

3.1 ELECTRICAL REQUIREMENTS

3.1.1 POWERTRAIN CONTROLLER MOUNTING BRACKET REQUIREMENTS

3.1.1.1 ECM

a. The ECM shall be Platform mounted.

b. Platform shall be responsible for the ECM mounting bracket.

c. GM Powertrain and Platform reviewed the ECM mounting location and design for MY08 (E38). The mounting was carryover through MY16 and will be for MY17 (E78).

3.1.1.2 TCM

a. The TCM shall be Powertrain mounted (T43 TCM is internal to the transmission, therefore shall be Powertrain mounted).

b. Powertrain shall be responsible for the TCM mounting bracket (T43 TCM is internal to the transmission, therefore Platform is not responsible for the mounting bracket).

c. GM Powertrain shall review the TCM mounting location and design for acceptance.

3.1.2 SYSTEM REQUIREMENTS

3.1.2.1 ENGINE WIRING HARNESS

a. Reference the engine controller IVED mechanization indicated in section 2.2.1.

b. Powertrain shall be responsible for any pre-wired engine harness.

3.1.2.2 HARNESS INSTALLATION AND REMOVAL

The ECM and TCM shall suffer no degradation in performance or inadvertent actuation while the battery voltage is in the ranges specified for vehicle battery and ignition in GMW8763 section 3.5, and when he following conditions occur:

(1) Disconnection and subsequent reconnection of vehicle battery.

(2) VBATT or ignition feed is connected before a ground connection is made.

(3) The installation or removal of any harness connectors, battery or ignition feed in any order or combination in all operating modes of ECM and TCM (Asleep, Ignition Run, etc.).

3.1.2.3 OVERLOAD PROTECTION

Reference PPEI for overload protection requirements. Refer to PPEI GM8763.

a. All inputs and outputs (I/O) of the ECM and TCM shall be protected from overloads resulting from shorts to battery or ground while VBATT is within any voltage range specified in GMW8763 section 3.5, and under all specified environmental conditions unless stated otherwise.

b. In the event of the loss of power or ground to the ECM or TCM, any “sneak path” caused by the parallel connection of an ECM or TCM input, to an input of another module that is also pulled up or pulled down, shall not result in any damage or permanent performance degradation of the ECM or TCM. In addition, such a sneak path shall not cause unintended actuation of any ECM or TCM function.

3.2 FUNCTIONAL REQUIREMENTS

This section defines the PPEI functional requirements for the engine control module (ECM) and the transmission control module (TCM).

3.2.1 ECM AND TCM REQUIREMENTS

The ECM and TCM shall meet the requirements of GM8762 to GM8781 v8.4 (PPEI) and shall meet any additional requirements for the vehicle configuration outlined in this document.

The files below show deviations requested from the PPEI requirements for the N-Series vehicle.

1. PPEI Deviation Request – Brake Pedal (modified MY16 implementation, not 100% PPEI 8.4 compliant):

PPEI Deviation Request #1 - Brake S

2. PPEI Deviation Request – Cruise Control (carry-over prior deviation)

PPEI Deviation Request #2 - Cruise

3. PPEI Deviation Request – Odometer calculation (carry-over prior deviation)

PPEI Deviation Request #3 - Odome

4. PPEI Deviation Request – No VTD, No Immobilizer (No 08MY VTD w/o immobilizer; GM vehicle detect)

PPEI Deviation Request #4 - VTD 24

5. PPEI Deviation Request – GMLAN content

   a. Wheel Speed messages (No PPEI 2.x Frame $280, No PPEI 8.4 $0C1/$0C5)

   b. Post collision operation

   c. Platform will not transmit $17D content required per CASPER feature to signal mapping activity (required for drive efficiency indication)

   d. PPEI, GMNAE Power Mode and Vehicle Electrical Power Moding Specification using GMLAN communication Strategy Deviation Request – GMLAN signal deviations

   e. Platform will not transmit $4E1 VIN required per CASPER feature to signal mapping activity

   f. Platform to transmit “Antilock Brake System Present” = True

PPEI Deviation Request #5 - GMLAN

6. PPEI Deviation Request – Cruise switch mechanization; “Switch Data Integrity” GMLAN signal

PPEI Deviation Request #6 - Cruise

7. PPEI Deviation Request – Ignition Switch Mechanization

PPEI Deviation Request #7 - Ignitio

The information in the PPEI Spreadsheet shall have the following meaning:

a. Each section of the PPEI Spreadsheet contains a row for deviations to Hardwire, Serial Data, Calibrations and Algorithms. Platform/Powertrain shall comply with the stated deviations, e.g., ICR or CEA.

b. The "Usage" column reflects if:
    a. The signal or calibration is required to be used (i.e., Platform Required or Powertrain Required)
    b. OR
    c. The signal or calibration is not required to be used (i.e., Platform Optional or Powertrain Optional).

c. The "X-mitter" column reflects if:
    a. The signal is required to be transmitted (i.e., Platform Required or Powertrain Required)
    b. OR
    c. The signal is not required and is sent with a default value (i.e., Platform Optional or Powertrain Optional).

d. Any signal originating from the ECM/TCM or the Gateway module shall be transmitted with a predefined "default" value if the signal is not used.

e. Physical device configuration requirements for serial data signals are defined in GM8772, Section 3.2.2.

3.2.2 GMLAN SIGNAL SUMMARY

Refer to PPEI Spreadsheet for GMLAN signal usage.

Refer to GMW8762 v8.4 for the GMLAN signal definition.

Refer to Platform GMLAN Data Dictionary. Reference appendix D

3.2.3 FEATURES USED

3.2.3.1 IDLE BOOST

TBD. For MY17, Isuzu will have to replace use of $285 message (no longer present).

3.3 TRACEABILITY DATA

The ECM and TCM shall meet the requirements defined in the Electronic Control Module Traceability Technical Specification (GMW4710).

3.4 DIAGNOSTIC REQUIREMENTS

a. The ECM and TCM shall meet the requirements defined in the Enhanced Diagnostic Test Mode Specification (GMW3110).

b. The ECM and TCM shall meet the requirements defined in the Service & Assembly Diagnostic Requirement Specification. Reference Appendix A.

3.5 FLASH PROGRAMMING

a. The ECM and TCM shall meet the requirements defined in the Enhanced Diagnostic Test Mode Specification (GMW3110).

3.5.1 BOOT SOFTWARE

a. The ECM and TCM boot software shall be designed to prevent an inoperable controller if an error occurs during a programming event.

   1) GM Platform shall review the ECM and TCM boot software reprogramming strategy for acceptance.

b. The erase and write routine should not reside in the ECM or TCM boot software.

3.5.2 CALIBRATION DEVELOPMENT SYSTEM

a. The ECM and TCM software and calibration files shall be CAL-DS compatible.

b. The ECM shall contain Powertrain calibration partition(s).

c. The ECM shall contain two Platform calibration partitions:

   1) Vehicle System Calibration.

   2) Vehicle Speed Calibration.

3.5.3 CALIBRATION FILES

The calibration files shall be address independent, with the calibration address located in the ECM or TCM software header.

3.5.4 BILL OF PROCESS

a. Powertrain shall have a total flash programming time of 60 seconds. For each platform, this time includes all Powertrain modules that will be programmed in the assembly plant.

b. The assembly plant shall only flash program calibration data files.

4.0 VALIDATION

4.1 COMPONENT TESTS

An ECM and TCM design validation (DV) plan, a product validation (PV) plan, and a continuing conformance (CC) plan shall be developed by GM Powertrain and their supplier with the approval of GM to meet the environmental and functional requirements. The supplier and GM Powertrain shall execute the plans to evaluate the performance, durability, reliability, and electromagnetic compatibility of the ECM and TCM.

4.2 VEHICLE TESTS

The ECM and TCM shall meet the corporate product performance objectives for the following Vehicle Level Test Procedures. The ECM and TCM shall operate per specification without damage, degradation in performance, or inadvertent output de/activation during the tests. The ECM and TCM shall not set any diagnostic trouble codes (DTC) during the tests unless such DTCs are valid faults.

Please refer to the GM/Isuzu SOW for Powertrain Engineering Services for 2017 MY Isuzu N-Series Gas Truck dated 04-December-2014.

APPENDIX A: SERVICE & ASSEMBLY DIAGNOSTIC REQUIREMENT SPECIFICATION

E78 ECM: Diagnostic Component Technical Specification

E78_diag_cts_1115B 11201.pdf

DiagTestModeConfigSpec-16.16.112.00

T43 TCM: Diagnostic Component Technical Specification

GMPT_T43_Diagnostic_CTS-Rev2p5.zip

PIM : Diagnostic Component Technical Specification

TBD

APPENDIX B: CALIBRATION RELATED CEA

The ICRs listed below are required for the N-Series program:

PPEI 8.4 ICR #	Description

APPENDIX C: CES PROGRAM IMPLEMENTATION

CES#	CES Description	Notes
??????	Initial CASPER Release	Rev 001

APPENDIX D: GMLAN DATA DICTIONARY

N-Series PPEI 8.4 Message List

Feature to Signal Mapping_Isuzu_17JI

GLOSSARY

4k-ppm	**4000 pulses per mile**
4WD	**Four Wheel Drive**
A	**Amperes**
ABS	**Antilock Braking System**
A/C	**Air Conditioning**
ALDL	**Assembly Line Data Link**
ATC	**Active Transfer Case**
AWD	**All Wheel Drive**
BASS	**Brake Applied Sensing System**
BOP	**Bill of Process**
°C	**Degrees Celsius**
CAL-DS	**Calibration Development System**
CAN	**Controller Area Network**
CASPER	**Custom Application Specific PPEI Electrical Requirements**
CC	**Continuing Conformance**
CEA	**Corporate Engineering Authorization**
CTR	**Component Technical Requirement**
CTS	**Component Technical Specification**
DIC	**Driver Information Center**
DTC	**Diagnostic Trouble Code**
DV	**Design Validation**
DVT	**Dynamic Vehicle Test**
ECM	**Engine Control Module**
E/E	**Electrical/Electronic**
EEPROM	**Electronic Erasable Programmable Read Only Memory**

EMC	**Electromagnetic Compatibility**
ESD	**Electrostatic Discharge**
ETC	**Electronic Throttle Control**
ETS	**Enhanced Traction System**
FMEA	**Failure Mode and Effects Analysis**
FWD	**Front Wheel Drive**
GM	**General Motors**
GME	**General Motors Europe**
GMLAN	**General Motors Local Area Network**
GMPT	**General Motors Powertrain**
GMW	**General Motors Worldwide Specification**
HSCAN	**High Speed Controller Area Network**
ICR	**Interface Change Request**
I/O	**Input / Output**
IP	**Instrument Panel**
IVED	**Integrated Vehicle Electrical Design**
LED	**Light Emitting Diode**
mA	**milliamperes**
mH	**millihenries**
mV	**millivolts**
NAPD	**North America Product Development**
OCS	**Oil Condition Sensor**
OLM	**Oil Life Monitor (software algorithm based)**
P/N	**Park/Neutral**
PCM	**Powertrain Control Module**
PPEI	**Platform-Powertrain Electrical Interface**

PV	**Product Validation**
PWM	**Pulse Width Modulated**
RAM	**Random Access Memory**
RDM	**Rear Drive Module**
RFA	**Remote Function Actuation**
RMS	**Root Mean Squared**
ROM	**Read Only Memory**
RVC	**Regulated Voltage Control**
RVS	**Remote Vehicle Start**
Sec	**Seconds**
SAD	**System Architecture Document**
SAE	**Society of Automotive Engineers**
SSTS	**Subsystem Technical Specification**
SSR	**Subsystem Requirements**
SRD	**Systems Requirements Document**
TBD	**To Be Determined**
TCM	**Transmission Control Module**
TCS	**Traction Control System (Full Functional)**
TEHCM	**Transmission Electro-Hydraulic Controller Module**
TUTD	**Transmission Tap-Up/Tap-Down**
V	**Volts**
Vbatt	**Vehicle battery**
VSES	**Vehicle Stability Enhancement System**
VSS	**Vehicle Speed Sensor**
VTD	**Vehicle Theft Deterrent**
VTS	**Vehicle Technical Specification**

CHANGE HISTORY

Date	Revision	Editor	Description	Section
8/22/2014	Rev. 1.0	S. Kekel	Initial Preliminary release	All
9/4/2014	Rev. 2.0	S. Kekel	Added note to section 2.2.1	2.2.1
			Added note to section 2.2.2	2.2.2
			Updated signal list ($1F1)	Appendix D
9/8/2014	Rev. 3.0	S. Kekel	Updated signal list ($0F1) for BTM	Appendix D
9/24/2014	Rev. 4.0	S. Kekel	Changed “supplier” to “parties to the Isuzu N-Series agreement”	1.1
			Schematic changes pending. Removed “remove BAS” pending change.	2.2.1
			Updated PPEI GenInfoElectricalSerialData. Re-add PPEI_PostCollisionOperation_REQ. Add PPEI_Transmission_REQ, PPEI_CoolingFanControl_REQ, and PPEI_SerialDataArchitecture_REQ. Updated note.	2.2.2
			Re-add #c.	3.1.1.1
			updated references to GMW3119 (previously updated to just “PPEI”)	3.1.2.2 3.1.2.3 3.2.1 3.2.2
			Changed “TBD” to “Please refer to the Isuzu N-Series agreement for the list of testing to be completed and the responsibilities of GM, Isuzu, and FEV.”	4.2
			Updated signal list $0F1. Add $0BE, $3C9, $3ED.	Appendix D
10/23/2014	Rev. 5.0	S. Kekel	Removed note regarding Isuzu modification of GMNAE Power Mode and Vehicle Electrical Power Moding Specification using GMLAN communication Strategy	2.2.3
			Added VTD deviation	3.2.1
			Added placeholder for TBD PPEI Deviation Request Cruise switch mechanization	3.2.1
			Added placeholder for TBD PPEI Deviation Request Post Collision Operation	3.2.1
			Added placeholder for TBD PPEI Deviation Request – Platform will not transmit $17D content required per CASPER feature to signal mapping activity (required for drive efficiency indication)	3.2.1

			Added PPEI, GMNAE Power Mode and Vehicle Electrical Power Moding Specification using GMLAN communication Strategy Deviation Request – $1F1 signal deviations	3.2.1
			Added placeholder TBD PPEI Deviation Request – Platform will not transmit $4E1 VIN required per CASPER feature to signal mapping activity	3.2.1
			Added PPEI Deviation Request – Platform to transmit "Antilock Brake System Present" = True	3.2.1
			Updates to Feature to Signal Mapping_Isuzu: • Add $2F9 to PIM for broadcast • $3F1 Generator Regulator Setpoint Duty Cycle Request value change • $4D1 Low Oil Pressure Warning Lamp typo fix • Update deviation section references in notes • Update some GMPT broadcast notes	Appendix D
			Added notes to ECM mech	2.2.1
06/03/2015	Rev. 6.0	S. Kekel	Implemented suggested edits provided by Isuzu 3/15/2015	various
			Multiple updates to Deviation section	3.2.1
6/24/2015	Rev. 6.1	S. Kekel	Updated deviations for consistency; re-added #1 (changed from reference)	3.2.1
			Updated section wording	3.2.1
			Updated PIM Spec to reference	2.2.2
			Updated signal list per Deviation #1	Appendix D